

DEVAANSH KUMAR

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CAREER PROFILE

- UNSW Computer Science graduate with professional software engineering experience across production full-stack development, backend systems, Python, TypeScript, React and AI-enabled applications
- Technical experience spanning software engineering, backend/API development, Python, AI/ML, computer vision, databases, networking, operating systems and algorithms
- Proven problem-solving, communication and leadership through production engineering, an industry-client software capstone, academic tutoring and student leadership

EDUCATION

Bachelor of Science (Computer Science)

September 2023 - September 2026

University of New South Wales

- UNSW International Student Excellence Award - A merit-based scholarship worth AUD 8,800
- UNSW Employability Award — 2026
- Active member of CSESoc, DevSoc and CSE Revue

TECHNICAL SKILLS

- Programming: Python, TypeScript, JavaScript, SQL, Java, C, C++, Rust, R, Bash/Shell
- Software & Backend: React, Next.js, Node.js, FastAPI, Express, REST APIs, WebSockets, Prisma, Drizzle ORM, OpenAPI
- Data & AI: PyTorch, Pandas, XGBoost, Machine Learning, Computer Vision, Regression & Classification, Data Visualisation
- Databases & Infrastructure: PostgreSQL, MySQL, Docker, Git, Linux, CI/CD, Vercel
- Testing & Engineering Tools: Pytest, Vitest, Postman, Jira, PostHog
- Tools & Platforms: Figma, Canva, Microsoft Office, Google Workspace
- Languages: English (Fluent), Hindi (Native)

EXPERIENCE

Software Engineer (Casual)

August 2026 - September 2026

Arms Operations Analysis Pty Ltd, Sydney

- Selected for paid follow-on engineering after an industry-client capstone to help mature the Autonomous Network Correction and Recovery (ANCR) concept demonstrator and consolidate multiple software prototypes into a unified system
- Engineered Python/FastAPI and React/TypeScript capabilities for network simulation, real-time monitoring, RF/network degradation analysis, recovery workflows and operator decision support
- Implemented cross-stack functionality spanning tactical user interfaces, predictive network-recovery workflows, recovery-drone simulation, REST/OpenAPI APIs, Docker deployment, automated testing and end-to-end system integration

Academic Tutor

March 2026 - Present

Shalom College UNSW, Sydney

- Deliver individual and small-group tutoring across computer science, mathematics, physics, accounting and finance
- Mentor 3–5 recurring university students weekly while providing additional academic support across a 130+ student residential community
- Strengthen conceptual understanding, problem-solving ability and academic confidence across first-year university coursework

Software Engineering Intern

October 2025 - March 2026

Tandem Learning, Sydney

- Developed and shipped 8–10 production features for a live education platform using Next.js, React, TypeScript, Prisma and Drizzle ORM within a 4–5 engineer product team
- Built and stabilised Benchmark Mode, developing multi-step assessment configuration, attempt and results workflows while resolving App Router, TypeScript, routing and production-build issues
- Improved the Wordsmith AI autograder, debugging prompt and grading-pipeline behaviour for visual questions to produce more reliable, evidence-grounded assessment feedback

- Refactored and extended Agent0 AI chat using the Vercel AI SDK, working across streamed AI responses, tool integrations, prompt architecture and client/server message handling
- Integrated product analytics and resolved production issues across frontend rendering, APIs, validation, database-backed workflows and CI/build pipelines on a platform supporting 1,642+ students and 8,026+ assessment attempts

Summer Resident Advisor

December 2025 - February 2026

Shalom College UNSW, Sydney

- Served as primary after-hours point of contact for 130+ residents, overseeing safety, operational continuity and incident response
- Managed overnight and weekend duty shifts requiring independent judgement, escalation management and time-critical decision-making
- Acted as Chief Fire Warden, coordinating emergency response procedures and incident management when required

Treasurer

October 2024 - February 2026

Shalom College UNSW, Sydney

- Managed budgeting, expense tracking and financial allocation for a \$35,000+ student budget supporting 130+ residents
- Delivered financial planning and governance across 20+ student events, improving budget efficiency and identifying \$2,000–\$4,000 in savings
- Collaborated with an 8-member student leadership team to manage operations, engagement and community initiatives

RELEVANT PROJECTS

Network Analytic Tool (NAT) - Industry Client Capstone

June 2026 - August 2026

UNSW Computer Science Project

- Architected and delivered a Dockerised Python/FastAPI and React/TypeScript network-resilience platform, integrating simulation, REST/OpenAPI APIs, interactive visualisation and mission-aware decision support
- Engineered end-to-end network simulation and recovery workflows spanning resilient communications, configurable RF degradation, network-health analytics, terrain-aware RF modelling, stress-event simulation and recovery-drone simulation
- Drove full-stack integration and production readiness across backend services, shared API contracts, frontend interoperability, Docker deployment, automated testing, CI/CD and system-wide debugging
- Led technical delivery within a five-person engineering team across Jira/Git workflows, client requirements and demonstrations, quality assurance and final handover, with comprehensive automated test coverage

Computer Vision – Wheat Crop Segmentation

February 2026 - May 2026

UNSW Computer Vision

- Developed and evaluated four semantic image-segmentation pipelines spanning classical computer vision, graph-based ML, clustering and deep learning
- Built a PyTorch workflow with U-Net and pretrained ResNet50 transfer learning, data augmentation and structured evaluation
- Achieved approximately 0.91 IoU / 0.95 F1, while analysing robustness, inference trade-offs and deployment considerations

BitTrickle – Peer-to-Peer File Sharing System

September 2024 - December 2024

UNSW Computer Network and Applications

- Designed and implemented a peer-to-peer file-sharing system in Python, using UDP for control messaging and TCP for direct peer-to-peer file transfer
- Designed a custom application-layer protocol supporting authentication, peer discovery, file publishing, search and on-demand retrieval
- Implemented heartbeat-based peer liveness tracking, concurrent file serving, multithreading and session-aware server-side state management

EXTRA CURRICULAR

- Harvard CS50's Introduction to Artificial Intelligence with Python, Harvard University (In Progress)
- Electronic Arts Software Engineering Virtual Experience, Forage (Jan 2025) – Proposed gameplay feature enhancements and practised software design using C++
- AWS APAC Solutions Architecture Virtual Experience, Forage (Jan 2025) – Designed scalable cloud architecture solutions with cost and performance considerations
- Harvard CS50's Introduction to Programming with Python, Harvard University (2022) – Completed 9 problem sets and a final project in Python
- HarvardX Technology Entrepreneurship: Lab to Market, Harvard University (2022) – Completed coursework on problem-solving, business architecture, operating models, and market readiness
- TechnoXian World Robotics Championship (2022) – Led a team in an international robotics competition